

returns and volatilities especially important; if you forecast lower volatility, you can spend more for a given level of returns.

This framework can be used to estimate sustainable withdrawal rates for a given failure rate and retirement duration for most any kind of capital market expectations. The purpose here is to demonstrate how retirees can translate their own expectations into an understanding about how to choose a withdrawal rate and asset allocation strategy.

Dynamic Asset Allocation Strategies

A second common assumption for withdrawal rate studies is that the asset allocation remains fixed throughout retirement and is rebalanced to the targeted allocation every year. For this discussion, we will relax the assumption that retirees maintain the same fixed asset allocation throughout their retirements, but we will remain firmly within the realm of total-return investing. In chapter 7, we will also consider time segmentation as a different way to justify a dynamic asset allocation in retirement.

Changing asset allocation over retirement within the context of a total-return investment portfolio can be understood as an attempt to reduce portfolio volatility at key points in retirement in order to reduce the retiree's exposure to sequence-of-returns risk. Even if the average market return is decent, retirees are especially vulnerable to the impact of bad market returns in early retirement. Sequence risk is uniquely caused by the attempt to spend a constant amount each year from a volatile investment portfolio. Sequence risk can be dampened by either letting spending fluctuate or reducing portfolio volatility. For now, we'll focus on reducing volatility.

Outside of just using a low-equity allocation throughout retirement (which comes with its own sets of risks in terms of being unable to support a spending goal beyond the bond yield curve), we can identify three major ways to reduce portfolio volatility when it counts the most. These methods have three goals:

1. To reduce volatility when most exposed to absolute wealth losses (rising equity glide path),
2. To reduce volatility when most exposed to "predictable" market losses (valuation-based asset allocation), or